PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 11 **– Abstraction and Interfaces**

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1. **Task11.md**

TASK 11 – Complete Campus Smart-Parking System

Aim

To implement and test a complete Campus Smart-Parking System using Java object-oriented programming concepts.

System Features

1. Supports cars, two-wheelers and electric vehicles.
2. Allocates available parking slots.
3. Generates unique parking tickets.
4. Records vehicle entry and exit.
5. Provides monthly passes for students and staff.
6. Calculates parking fees for visitors.

OOP Concepts Used

- Classes and Objects
- Constructors
- Encapsulation
- Inheritance
- Method Overriding
- Runtime Polymorphism
- Abstraction

Testing

The system is tested with different vehicle types, parking allocation, ticket generation, monthly passes, visitor payments and unavailable slots.

Conclusion

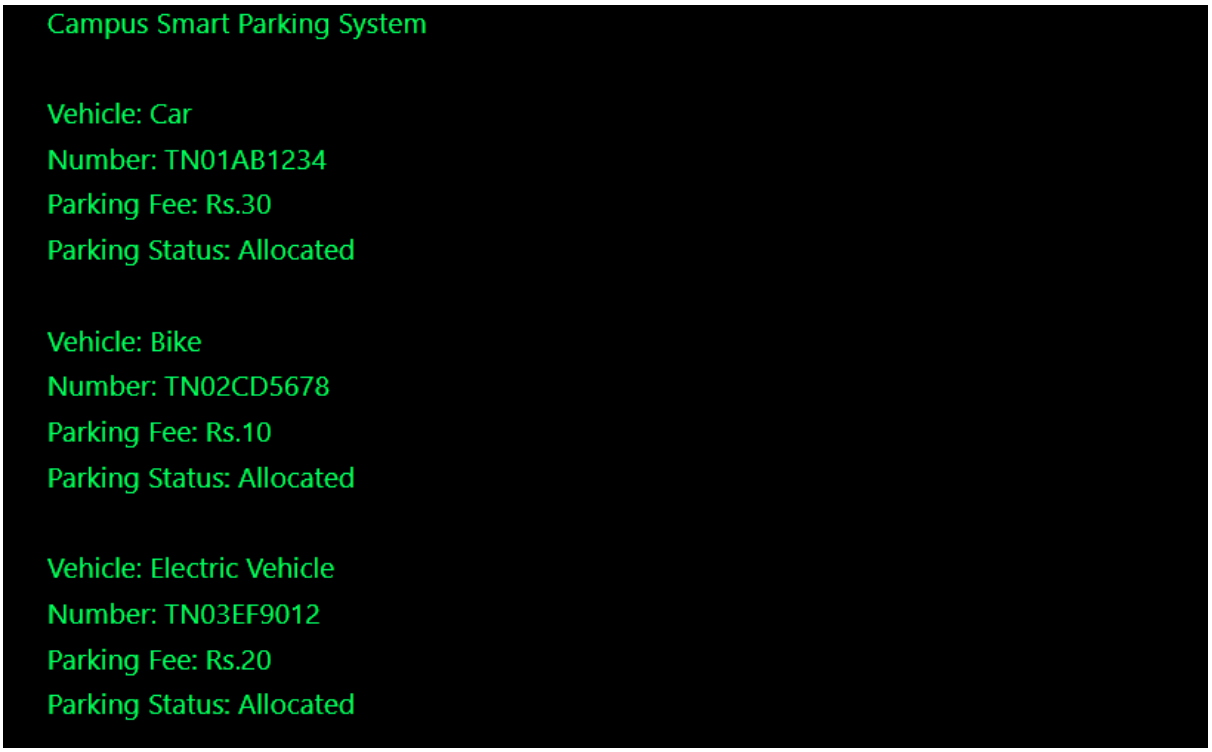
The Campus Smart-Parking System provides an organized way to manage parking spaces, vehicle entry and exit, tickets and parking payments using Java.

2. CampusParkingSystem.java

```
class Vehicle {
    String number;
    Vehicle(String number) {
        this.number = number;
    }
}
public class CampusParkingSystem {
    public static void main(String[] args) {
        Vehicle[] vehicles = {
            new Vehicle("TN01AB1234"),
            new Vehicle("TN02CD5678"),
            new Vehicle("TN03EF9012")
        };
        String[] types = {"Car", "Bike", "Electric Vehicle"};
        int[] fees = {30, 10, 20};
        System.out.println("Campus Smart Parking System");
        for (int i = 0; i < vehicles.length; i++) {
            System.out.println("\nVehicle: " + types[i]);
            System.out.println("Number: " + vehicles[i].number);
            System.out.println("Parking Fee: Rs." + fees[i]);
            System.out.println("Parking Status: Allocated");
```

```
        }
    }
}
```

3. Output:


Campus Smart Parking System
Vehicle: Car
Number: TN01AB1234
Parking Fee: Rs.30
Parking Status: Allocated
Vehicle: Bike
Number: TN02CD5678
Parking Fee: Rs.10
Parking Status: Allocated
Vehicle: Electric Vehicle
Number: TN03EF9012
Parking Fee: Rs.20
Parking Status: Allocated